

# 1<sup>st</sup> Law of Thermodynamics

The following table gives data, in kJ, for a system undergoing a thermodynamic cycle consisting of four processes in series. For the cycle, kinetic and potential energy effects can be neglected. Determine:

- The missing table entries, each in kJ.
- Whether the cycle is a power cycle or a refrigeration cycle?

[PU 2010 Fall]

| Process | $\Delta U$ | Q   | W     |
|---------|------------|-----|-------|
| 1-2     | 600        |     | -600  |
| 2-3     |            |     | -1300 |
| 3-4     | -700       | 0   |       |
| 4-1     |            | 500 | 700   |

- Steam enters a horizontally aligned well-insulated nozzle of inlet area  $0.15\text{m}^2$  at the velocity of  $65\text{m/s}$ . At the inlet, the specific enthalpy of steam is  $3000\text{kJ/kg}$ , specific volume  $0.187\text{m}^3/\text{kg}$  and at the outlet specific enthalpy is  $2762\text{kJ/kg}$ , specific volume  $0.498\text{m}^3/\text{kg}$ . Determine:

- Velocity of the steam at the outlet
- Mass flow rate of steam
- Exit area of nozzle

[PU 2010 Fall]

- The properties of a system, during a reversible constant pressure non-flow process at  $P=1.6\text{ bar}$ , changes from  $V_1=0.3\text{m}^3/\text{kg}$ ,  $T_1=20^\circ\text{C}$  to  $V_2=0.55\text{m}^3/\text{kg}$ ,  $T_2=260^\circ\text{C}$ . The specific heat of the fluid is given by

$$C_p = 1.5 + 75/(T+45)\text{kJ/kg}^\circ\text{C} \text{ where } T \text{ is in } ^\circ\text{C}.$$

Determine:

- Heat added /kg
- Change in internal energy/kg
- Change in enthalpy/kg

[PU 2009 Fall]

- Nitrogen is heated while it flows steadily through a constant area tube. At the inlet the nitrogen is  $200\text{ kPa}$ ,  $100^\circ\text{C}$ , and has a velocity of  $175\text{ m/s}$ . It is heated to  $300^\circ\text{C}$ . At the outlet pressure is  $115\text{ kPa}$ . At the inlet,  $u_1 = 227\text{ kJ/kg}$  and  $h_1 = 387\text{ kJ/kg}$ ; at the exit  $u_2 = 428\text{ kJ/kg}$  and  $h_2 = 598\text{ kJ/kg}$ . Determine the heat transfer per kg of nitrogen. [PU2009 Fall/2008 Spring]

- A turbine operating under steady flow conditions receives  $5000\text{kg}$  of steam per hour. The steam enters the turbine at a velocity of  $3000\text{ m/min}$  at an elevation of  $5\text{m}$  and a specific enthalpy of  $278\text{ kJ/kg}$ . It leaves the turbine at a velocity of  $6000\text{ m/min}$ , at an elevation of  $1\text{m}$  and a specific enthalpy of  $2259\text{ kJ/kg}$ . Heat losses from the turbine to the surrounding amounts to  $16736\text{ kJ/kg}$ . Determine the power output of the turbine. [PU 2009 Spring]

- $1\text{ kg}$  of air can be expanded between two states as it does  $20\text{ kJ}$  of work and receives  $16\text{ kJ}$  of heat. A second kind of expansion can be found between the same initial and final states which require a heat input of only  $9\text{ kJ}$ . What is the change of internal energy in the first expansion and what is the work done by the air in the second expansion? [PU 2004 Spring]

- Steam enters a turbine at a pressure of 150 kPa and velocity of 200 m/s. The steam leaves the turbine at a pressure of 150 kPa and velocity of 200 m/s. Determine the work per kg of steam flowing through the turbine, assuming the process to be reversible and adiabatic. [PU 2004 Spring]
8. Steam enters a convergent divergent nozzle with a velocity of 60 m/s and 800 kPa and 300°C. The steam leaves the nozzle at 160 kPa and 0.96 dryness. The cross sectional area at the exit is 12 cm<sup>2</sup> and considering the flow as adiabatic, determine the steam velocity at the exit and steam flow rate. [PU 2005 Fall]
9. Steam at 0.6 MPa, 200°C enters an insulated nozzle with a velocity of 50 m/s. It leaves at a pressure of 0.15 MPa and a velocity of 60 m/s. Determine the final temperature if the steam is superheated. [PU 2006 Spring]
10. A steam turbine receives steam from the boiler at the rate of 5 kg/s and at 3.5 MPa and in dry saturated condition. The velocity of steam entering the turbine is 75 m/s and the elevation at the entry is 5 m. It leaves the turbine at a pressure of 7 kPa and with a dryness fraction of 0.8. The velocity of steam at the exit of the turbine is 160 m/s and elevation at the exit is 1 m. The heat loss from the turbine to the surrounding is 18 MJ/hr. Calculate the power developed by the turbine. [PU 2006 Spring]
11. Steam enters the nozzle at 1 MPa, 300°C, with a velocity of 30 m/s. the pressure of the steam at the nozzle exit is 0.3 MPa. Determine the exit velocity of the steam from the nozzle, assuming a reversible, adiabatic, steady state and steady flow process. [PU 2006 Fall]
12. A certain water heater operates under steady flow conditions receiving 4.2 kg/s of water at 75°C temperature, enthalpy 313.93 kJ/kg. The water is heated by mixing with steam which is supplied to the heater at temperature 100.2°C and enthalpy 2676 kJ/kg. The mixer leaves the heater as liquid water at temperature 100°C and enthalpy 419 kJ/kg. How much steam must be supplied to the heater per hour? [PU 2006 Fall]
13. Steam enters a turbine operating at steady state with a mass flow rate of 4600kg/h. The turbine develops a power output of 1000kW. At the inlet, the pressure is 60. bar, the temperature is 400°C, and velocity is 10 m/s. At the exit, the pressure is 0.1 bar, the quality is 0.9 and the velocity is 50 m/s. Calculate the rate of heat transfer between the turbine and the surroundings, in kW. [2007 Fall]
14. A steam turbine produces 500 MW of power. The inlet and outlet conditions of steam are as follows:

|             | Inlet    | Outlet  |
|-------------|----------|---------|
| Pressure    | 12.5 MPa | 10kPa   |
| Temperature | 500°C    | -       |
| Quality     | -        | 0.85    |
| Velocity    | 50m/s    | 100 m/s |

- Heat transferred to the surrounding from turbine surface is estimated as 10 MW. What is the mass flow rate of steam? [PU 2007 Spring]
15. Air undergoes a cyclic process in a cylinder and piston arrangement. First the atmospheric air is at 1 bar and 27°C is compressed adiabatically to 10 bar, then expanded isothermally up to initial pressure, and then brought to initial condition under constant pressure. Find
- i] The change in internal energy



16. A gas is held in a cylinder behind a piston undergoes a cycle composed of three processes:  
 Process 1-2: an adiabatic compression process. For this process  $W_{1,2} = -58 \text{ kJ}$   
 Process 2-3: an expansion during which heat is added and work is done on the piston.  
 $Q_{2,3} = 253 \text{ kJ}$ ;  $W_{2,3} = 95 \text{ kJ}$   
 Process 3-1: a cooling of the gas while the piston is stationary to bring the gas to its initial state.

Determine the net heat transfer and heat transfer of process 3-1 [PU 2008 Spring]

17. The internal energy of a certain substance is given by the following equation:

$$u = 3.56pv + 84, \text{ where } u \text{ is given in kJ/kg, } p \text{ is kPa, and } v \text{ is in m}^3/\text{kg}.$$

A system composed of  $\frac{1}{2} \text{ kg}$  of this substance expands from an initial pressure of 500 kPa and a volume of  $0.22 \text{ m}^3$  to a final pressure 100 kPa in a process in which pressure and volume are related by  $pv^{1.2} = \text{const.}$

- (a) If the expansion is quasi-static, find  $Q$ ,  $\Delta U$ , and  $W$  for the process  
 (b) In another process the same system expands according to the same pressure- volume relationship as in part (a), and from the same initial state to the same final state as in part (a), but the heat transfer in this case is 30 kJ. Find the work transfer for this process.  
 (c) Explain the difference in work transfer in parts (a) and (b).
18. A stationary fluid system goes through a cycle shown in figure comprising the following processes:

- (i) Process 1-2; isochoric heat addition of 235 kJ/kg  
 (ii) Process 2-3; adiabatic expansion to its original pressure with loss of 70 kJ/kg in internal energy.  
 (iii) Process 3-1; isobaric compression to its original volume with heat rejection of 200 kJ/kg.

Prepare a balance sheet of energy quantities and find the overall changes during the cycle.

19. The properties of certain fluid are related as follows:

$$u = 196 + 0.718t \text{ and } pv = 0.287(t + 273)$$

where  $u$  is the specific internal energy (kJ/kg),  $t$  is in  $^{\circ}\text{C}$ ,  $p$  is pressure (kN/m $^2$ ), and  $v$  is specific volume (m $^3$ /kg). For this fluid, find  $c_v$  and  $c_p$ .

### Answers:

- 0; 300; -1000; 700; -200; refrigeration cycle
- 692.98 m/s; 52.1 kg/s; 0.0375 m $^2$
- 475.94 kJ; 435.94 kJ; 475.94 kJ
- 274.95 kJ/kg; 247.97 kJ
- 5
- 13 kJ

8.  $188.25^{\circ}\text{C}$

9.  $188.37^{\circ}\text{C}$ ,  $185.86^{\circ}\text{C}$

10. 3.52 MW

11. 735.54 m/s

12. 703.88 kg/h

13. -61.35 kW

14. 458.52 kg/s

15. i) 200 kJ/kg; -200 kJ/kg; 0; ii) 297; 0; -281.44; 0 iii) 0; 198.25; -280; -82.25

iv) -200 kJ/kg; 198 kJ/kg; -80 kJ/kg; -82.25 kJ/kg

16. 37 kJ; -216 kJ

17. a) 36.5 kJ; -91 kJ; 127.5 kJ b) 121 kJ

18.  $\sum Q = \sum W_{\text{cycle}} = 35 \text{ kJ/kg}$

19. 0.718; 1.005 kJ/kgK

# 1st law of thermodynamics

① (i) The missing table entries, each in kJ.

| Process | $\Delta U$ | $Q$   | $W$   |
|---------|------------|-------|-------|
| 1-2     | 600        | 0     | -800  |
| 2-3     | 300        | -1000 | -1300 |
| 3-4     | -700       | 0     | 700   |
| 4-1     | -200       | 500   | 700   |

→ sol<sup>n</sup>:-

For process 1-2,  $Q = \Delta U + W$

$$\text{or, } 0 = 600 - 800$$

$$\therefore Q = 0$$

For process 3-4,  $Q = \Delta U + W$

$$\text{or, } 0 = -700 + W$$

$$\therefore W = 700$$

For process 4-1,  $Q = \Delta U + W$

$$\text{or, } 500 = \Delta U + 700$$

$$\therefore \Delta U = -200$$

For a cyclic process,  $\oint \Delta U = 0$

$$\text{i.e. } 600 + X - 700 - 200 = 0$$

$$\text{or, } 600 + X - 900 = 0$$

$$\therefore X = 300$$

For process 2-3,  $Q = \Delta U + W$

$$= 300 - 1300$$

$$\therefore Q = -1000$$

(ii) sol<sup>n</sup>:-

$$\begin{aligned}\text{Total work output} &= -800 - 1300 + 700 + 700 \\ &= -500 \text{ kJ}\end{aligned}$$

Since, the total work output is negative. So, the cycle is refrigeration cycle.

Q. sol<sup>n</sup> :-

At inlet,

$$\text{Area } (A_1) = 0.15 \text{ m}^2$$

$$\text{velocity } (V_1) = 65 \text{ m/s}$$

$$\text{specific enthalpy } (h_1) = 3000 \text{ kJ/kg}$$

$$= 3000 \times 10^3 \text{ J/kg}$$

$$\text{specific volume } (V_1) = 0.187 \text{ m}^3/\text{kg}$$

At outlet,

$$\text{specific enthalpy } (h_2) = 2762 \text{ kJ/kg} = 2762 \times 10^3 \text{ J/kg}$$

$$\text{specific volume } (V_2) = 0.498 \text{ m}^3/\text{kg}$$

Q. sol<sup>n</sup> :-

Applying Stea SFEE to inlet and outlet of nozzle, we get

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

$$3000 \times 10^3 + \frac{65^2}{2} = 2762 \times 10^3 + \frac{V_2^2}{2}$$

$$V_2^2 = 480225$$

$$V_2 = 692.98 \text{ m/s}$$

Q. sol<sup>n</sup> :-

$$\text{mass flow rate of steam } (\dot{m}) = \frac{A_1 V_1}{V_1} = \frac{0.15 \times 65}{0.187}$$

$$= 52.13 \text{ kg/s}$$

50



iii) sol<sup>n</sup>:-

For mass balance,  $\frac{A_1 V_1}{V_1} = \frac{A_2 V_2}{V_2}$

or,  $\frac{0.15 \times 65}{0.187} = \frac{A_2 \times 692.98}{0.498}$

or,  $A_2 = \frac{9.75 \times 0.498}{0.187 \times 692.98}$

or,  $A_2 = \frac{4.8555}{129.58}$

$\therefore A_2 = 0.0375 \text{ m}^2$

$\therefore$  Exit area of nozzle is  $0.0375 \text{ m}^2$ .

③ sol<sup>n</sup>:-

we have,  $c_p = 1.5 + \frac{75}{T+45} \text{ kJ/kg}^\circ\text{C}$

Also,  $P_1 = 1.6 \text{ bar} = 1.6 \times 10^5 \text{ Pa} = 1.6 \times 10^2 \text{ KPa} = 160 \text{ KPa}$

$V_1 = 0.3 \text{ m}^3/\text{kg}$

$T_1 = 20^\circ\text{C} = 20 + 273 = 293 \text{ K}$

$V_2 = 0.55 \text{ m}^3/\text{kg}$

$T_2 = 260^\circ\text{C} = 260 + 273 = 533 \text{ K}$

Then,

(i)  $c_p = \int_{T_1}^{T_2} c_p = \int_{T_1}^{T_2} \left( 1.5 + \frac{75}{T+45} \right) dT$

$= 1.5 + 75 \int_{293}^{533} \frac{dT}{T+45}$

$= 1.5 + 75 \left[ \ln(T+45) \right]_{293}^{533}$

$= 1.5 + 75 \left[ \ln(533+45) - \ln(293+45) \right]$

sol<sup>n</sup>:-

Here,  $P = 1.6 \text{ bar} = 1.6 \times 10^5 \text{ Pa} = 1.6 \times 10^2 \text{ kPa}$   
 $v_1 = 0.3 \text{ m}^3/\text{kg}$        $T_1 = 20^\circ\text{C} = 20 + 273 = 293\text{K}$   
 $v_2 = 0.55 \text{ m}^3/\text{kg}$        $T_2 = 260^\circ\text{C} = 260 + 273 = 533\text{K}$   
Then,

Heat added per kg:

We have total heat added = total change in enthalpy  
Also,  $C_p = \frac{dH}{dT}$

$$\therefore dH = C_p dT \Rightarrow Q_{1-2} = \int C_p dT \text{ (for const}^n \text{ pressure process)}$$

Integrating both sides w.r. to  $T$ ,

$$Q_{1-2} = H_2 - H_1 = \int_{T_1}^{T_2} C_p dT$$

$$= \int_{T_1}^{T_2} \frac{1.5T + 75}{T + 45} dT$$

$$= 1.5 \left[ T \right]_{T_1}^{T_2} + 75 \left[ \ln(T + 45) \right]_{T_1}^{T_2}$$

$$= 1.5 [260 - 20] + 75 [\ln(260 + 45) - \ln(20 + 45)]$$

$$= 1.5 [T]_{20}^{260} + 75 [\ln(T + 45)]_{20}^{260}$$

$$= 1.5 (260 - 20) + 75 [\ln(260 + 45) - \ln(20 + 45)]$$

$$= 360 + 75 [\ln 305 - \ln 65]$$

$$= 475.84 \text{ kJ/kg}$$

Work done during constant pressure non-flow process  
 $W_{1-2} = P(v_2 - v_1) = 1.6 \times 10^2 \times (0.55 - 0.3) = 40 \text{ kJ/kg}$



(ii) change in internal energy/kg

→ From 1st law of thermodynamics, we have

$$Q_{1-2} = U_2 - U_1 + W_{1-2}$$

$$\begin{aligned}\Rightarrow U_2 - U_1 &= Q_{1-2} - W_{1-2} \\ &= 475.94 \times 10^3 - 40000 \\ &= 435.94 \text{ kJ} \\ &= 435.94 \text{ kJ/kg}\end{aligned}$$

(iii) change in enthalpy/kg

→ since, the change in enthalpy in a constant pressure process is equal to the change in enthalpy, we have

$$\begin{aligned}H_2 - H_1 &= Q_{1-2} \\ &= 475.94 \\ &= 475.94 \text{ kJ/kg}\end{aligned}$$

so,

At inlet,

$$P_1 = 200 \text{ kPa}$$

$$T_1 = 100^\circ\text{C}$$

$$V_1 = 175 \text{ m/s}$$

$$u_1 = 227 \text{ kJ/kg}$$

$$h_1 = 387 \text{ kJ/kg}$$

$$q = ?$$

Then,

Applying SFEE to the inlet & outlet, we have,

$$\dot{m} \left( h_1 + \frac{V_1^2}{2} + g z_1 \right) + \dot{Q} = \dot{m} \left( h_2 + \frac{V_2^2}{2} + g z_2 \right) + \dot{W}_x$$

$$h_1 + \frac{V_1^2}{2} + g z_1 + q = h_2 + \frac{V_2^2}{2} + g z_2 + w_x$$

We put  $w_x = 0$ ,  $z_1 = z_2$

$$h_1 + \frac{V_1^2}{2} + q = h_2 + \frac{V_2^2}{2} \quad \text{--- (i)}$$

Here, mass flow rate is same

$$\text{i.e. } \dot{m}_1 = \dot{m}_2$$

$$\frac{A_1 V_1}{V_1} = \frac{A_2 V_2}{V_2} \quad (\because \text{Uniform area tube})$$

$$\text{or, } \frac{V_1}{V_2} = \frac{V_1}{V_2} \quad \text{--- (ii)}$$

$$\text{or, } \frac{V_1}{V_2} = \frac{V_1}{V_2}$$

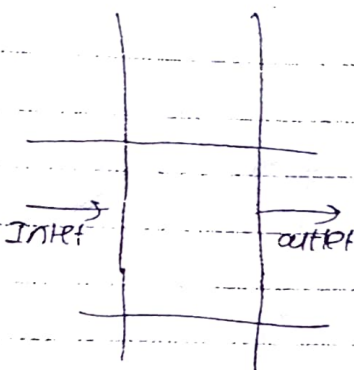
$$\text{or, } 175$$

At outlet,

$$h_2 = u_2 + P_2 v_2$$

$$\text{or, } 598 = 928 + 115 \times v_2$$

$$v_2 = 1.47 \text{ m}^3/\text{kg}$$



⑥ sol<sup>n</sup>:-

here,

for 1<sup>st</sup> expansion (A),

$${}_1W_2 = 20 \text{ kJ} \quad \& Q = 16 \text{ kJ}$$

For 2<sup>nd</sup> expansion (B),

$$\& Q = 9 \text{ kJ}$$

Then, from 1<sup>st</sup> law of thermodynamics, we have,

$$\& Q = U_2 - U_1 + {}_1W_2$$

For 1<sup>st</sup> expansion,  $16 = U_2 - U_1 + 20 \text{ kJ}$

$$\therefore U_2 - U_1 = -4 \text{ kJ}$$

For 2<sup>nd</sup> expansion,  $9 = \Delta U + {}_1W_2$

$$9 = -4 + {}_1W_2$$

$$\therefore {}_1W_2 = 13 \text{ kJ}$$



⑦ sol<sup>n</sup>:-

At inlet,

$$P_1 = 1 \text{ MPa} = 1 \times 10^6 \text{ Pa} = 10^3 \text{ kPa}$$

$$t_1 = 300^\circ \text{C} = 300 + 273 = 573 \text{ K}$$

$$V_1 = 50 \text{ m/s}$$

At outlet

$$P_2 = 150 \text{ kPa} = 150 \times 10^3 \text{ Pa}$$

$$V_2 = 200 \text{ m/s}$$

Then,

$$W_x = ?$$

Now,



From SFEE based on unit mass, we know that,

$$h_1 + \frac{v_1^2}{2} + gz_1 + q = h_2 + \frac{v_2^2}{2} + gz_2 + w$$

Assuming no change in P.E. and the process to be adiabatic i.e.  $q=0$ , then above equation becomes

$$h_1 + \frac{v_1^2}{2} = h_2 + \frac{v_2^2}{2} + w \quad \text{--- (i)}$$

To determine  $w$ , we have to determine values of  $h$  at inlet & outlet i.e.  $h_1$  and  $h_2$ .

At inlet : At 1000 kPa,

from steam table,  $t_s = 179.92^\circ\text{C}$

$t_1 = 300^\circ\text{C}$

Since,  $t_1 > t_s$ , the steam at inlet must be superheated.

Now, from steam table, we have

$h_1 = 3050.6 \text{ kJ/kg}$        $s_1 = 7.1219 \text{ kJ/kgK}$

Since, the process is reversible and adiabatic, it must be isentropic i.e.

$s_2 = s_1 = 7.1219 \text{ kJ/kgK}$

So, at outlet,  $p_2 = 150 \text{ kPa}$

$s_2 = 7.1219 \text{ kJ/kgK}$

But at  $p_2 = 150 \text{ kPa}$ ,

from steam table,

$s_f = 1.4338 \text{ kJ/kgK}$

$h_f = 467.18 \text{ kJ/kg}$

$s_{fg} = 5.7894 \text{ kJ/kgK}$

$h_{fg} = 2226.2 \text{ kJ/kg}$

$s_g = 7.2232 \text{ kJ/kgK}$

$h_g = 2693.4 \text{ kJ/kg}$

Since,  $s_f < s_2 < s_g$ , the steam at the outlet must be wet steam. Let,  $x_2$  be its dryness fraction.

$$\therefore s_2 = s_{f/p_2} + x_2 s_{fg/p_2}$$

$$\text{or, } 7.1219 = 1.4338 + x_2 \times 5.7894$$

$$\therefore x_2 = 0.9825$$

similarly, we have

$$h_2 = h_{f/p_2} + x_2 h_{fg/p_2}$$

$$\text{or, } h_2 = 467.18 + 0.9825 \times 2226.2$$

$$\therefore h_2 = 2654.42 \text{ kJ/kg}$$

so, from eq<sup>n</sup> (i), we have

$$\text{or, } 3050.6 \times 10^3 + \frac{50^2}{2} = \frac{2654.42 \times 10^3}{2} + \frac{200^2}{2} \text{ kJ}$$

$$\text{or, } w = 3051850 - 2674120$$

$$\text{or, } w = 377430 \text{ W}$$

$$\therefore w = 377.61 \text{ kW/kg}$$

⑧ Sol<sup>n</sup>:-

At inlet

$$V_1 = 60 \text{ m/s}$$

$$P_1 = 800 \text{ kPa} = 800 \times 10^3 \text{ Pa}$$

$$T_1 = 300^\circ \text{C} = 300 + 273 = 573 \text{ K}$$

At outlet,

$$P_2 = 160 \text{ kPa} = 160 \times 10^3 \text{ Pa}$$

$$x_2 = 0.96$$

$$A_2 = 12 \text{ cm}^2 = 1.2 \times 10^{-3} \text{ m}^2$$

then,  $V_2 = ?$  and  $\dot{m} = ?$



Applying SFEE based on unit mass, we have

$$\dot{m} \left( h_1 + \frac{V_1^2}{2} + gz_1 \right) + q = \dot{m} \left( h_2 + \frac{V_2^2}{2} + gz_2 \right) + \dot{W}$$

Assuming no change in P.F and the process to be adiabatic i.e.  $q = 0$ , then above eq<sup>n</sup> becomes

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \quad \text{--- (i)}$$

At 800 kPa,

from steam table,  $t_s = 170.44^\circ\text{C}$

$t_1 = 300^\circ\text{C}$

Since  $t_1 > t_s$ , so, the steam at inlet must be superheated one.

Now, from steam table,  $h_1 = 3059.9 \text{ kJ/kg}$

$s_1 = 7.2319 \text{ kJ/kgK}$

At 160 kPa, the properties of steam can not be obtained directly. so, it has to be determined by interpolation as follows:

| Pressure (kPa) | $h_f$ (kJ/kg) | $h_{fg}$ (kJ/kg) |
|----------------|---------------|------------------|
| 800 150        | 467.18        | 2226.2           |
| 160            | ?             | ?                |
| 175            | 487.08        | 2213.3           |

Interpolating for  $h_f$ , we get,

$$h_f - 467.18 = \frac{487.08 - 467.18}{175 - 150} (160 - 150)$$

$$h_f - 467.18 = 7.96$$

$$h_f = 475.14 \text{ kJ/kg}$$

$$h_f = 475.14 \text{ kJ/kg}$$



similarly, interpolating. for  $h_{fg}$ , we get,

$$h_{fg} = 477$$

$$h_{fg} - 2226.2 = \frac{2700.4 - 2226.2}{487.08 - 467.28} (477.08 - 467.28)$$

$$h_{fg} - 2226.2 = \frac{2213.3 - 2226.2}{175 - 150} (150 - 150)$$

$$\text{or, } h_{fg} - 2226.2 = -227 + 5.16$$

$$\therefore h_{fg} = 2453.3 \text{ kJ/kg} \quad 2221.04 \text{ kJ/kg}$$

so, finally at 160 kPa

$$h_{f/P_2} = 477.08 \text{ kJ/kg} \quad 475.14 \text{ kJ/kg}$$

$$h_{fg/P_2} = 2463.3 \text{ kJ/kg} \quad 2221.04 \text{ kJ/kg}$$

$$x_2 = 0.96$$

$$\therefore h_2 = h_{f/P_2} + x_2 \cdot h_{fg/P_2}$$

$$= 477.08 + 0.96 \times 2463.3$$

$$\therefore h_2 = 2841.848 \text{ kJ/kg}$$

$$\text{or, } h_2 = 475.14 + (0.96 \times 2221.04)$$

$$\therefore h_2 = 2607.33 \text{ kJ/kg}$$

$$\text{Then, } \left( 3055.9 \times 10^3 + \frac{60^2}{2} \right) = 2607.33 \times 10^3 + \frac{V_2^2}{2}$$

$$\text{or, } (3057.700 - 2607.33 \times 10^3) \times 2 = V_2^2$$

$$\text{or, } V_2^2 = 900740$$

$$\therefore V_2 = 949.07 \text{ m/s}$$

(i) free velocity of steam at exit = 949.07 m/s

$$[ \text{(ii) steam flow rate (m)} = \frac{A_2 \cdot V_2}{V_2} = \frac{12 \times 10^{-4} \times 949.07}{V_2} ]$$

(should be calculated with same process for  $h_f$  and  $h_{fg}$ )

Sol<sup>n</sup>:-

At inlet,  $P_1 = 0.6 \text{ MPa} = 0.6 \times 10^6 \text{ Pa} = 0.6 \times 10^3 \text{ kPa}$

$$V_1 = 50 \text{ m/s}$$

$$t_1 = 200^\circ\text{C}$$

At outlet,  $P_2 = 0.15 \text{ MPa} = 0.15 \times 10^6 \text{ Pa} = 0.15 \times 10^3 \text{ kPa}$

$$V_2 = 60 \text{ m/s}$$

$$t_2 = ?$$

Applying SFEE to the nozzle,

$$\dot{m} \left( h_1 + \frac{V_1^2}{2} + g z_1 \right) + \dot{q} = \dot{m} \left( h_2 + \frac{V_2^2}{2} + g z_2 \right) + \dot{w}$$

Since, the nozzle is insulated so,  $\dot{q} = 0$  &  $\dot{w} = 0$

Also, neglecting change in P.E, we have,

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \quad \text{--- (i)}$$

At 600 kPa, from steam table,

$$t_s = 158.86^\circ\text{C} \text{ and } t_1 = 200^\circ\text{C}$$

Since,  $t_1 > t_s$ . The steam is superheated one.

$$\therefore h_1 = 2849.7 \text{ kJ/kg}$$

Now, in eq<sup>n</sup> (i),

$$2849.7 \times 10^3 + \frac{50^2}{2} = h_2 + \frac{60^2}{2}$$

$$h_2 = 2849150 \text{ J/kg} = 2849.15 \text{ kJ/kg}$$

Then, At 150 kPa, from steam table,

$$h_f = 467.18 \text{ kJ/kg}$$



Q10) Sol<sup>n</sup>:-

At inlet condition

$$\dot{m} = 5 \text{ kg/s}$$

$$P_1 = 3.5 \text{ MPa} = 3.5 \times 10^6 \text{ Pa} \\ = 3.5 \times 10^3 \text{ kPa}$$

$$V_1 = 75 \text{ m/s}$$

$$z_1 = 5 \text{ m}$$

$$q = 18 \text{ MJ/hr} = 18 \times 10^6 \text{ J/hr}$$

$$= \frac{18 \times 10^6}{3600} = 5000 \text{ J/s} = 5 \text{ kJ/s}$$

At outlet condition

$$P_2 = 7 \text{ kPa} = 7 \times 10^3 \text{ Pa}$$

$$x_2 = 0.8$$

$$V_2 = 160 \text{ m/s}$$

$$z_2 = 1 \text{ m}$$

Applying SFEE to the turbine, or

$$\dot{m} \left[ h_1 + \frac{V_1^2}{2} + g z_1 \right] + q = \dot{m} \left[ h_2 + \frac{V_2^2}{2} + g z_2 \right] + \dot{W}_x$$

or, At inlet at  $3.5 \times 10^6 \text{ Pa} = 3500 \text{ kPa}$

$$h_g = 1049.6 \text{ kJ/kg}, 2801.7 \text{ kJ/kg}$$

$$\Rightarrow h_g - h_f = \frac{1049.6}{2801.7} \text{ kJ/kg (as it is dry saturated steam)}$$

Similarly,

at outlet, At ~~300~~ 7 kPa,  $h_f = 163.35 \text{ kJ/kg}$

$$h_{fg} = 2408.3 \text{ kJ/kg}$$

$$\therefore h_2 = h_f + x_2 \cdot h_{fg}$$

$$= 163.35 + (0.8 \times 2408.3)$$

$$= 2089.99 \text{ kJ/kg}$$

Then, in eq<sup>n</sup> (i) becomes,

$$5 \left[ \frac{1049.6}{10^3} + \frac{75^2}{2} + 9.8 \times 5 \right] + (5 \times 10^3) = 5 \left[ \frac{2089.99}{10^3} + \frac{160^2}{2} + 9.8 \times 1 \right] + \dot{W}_x$$

$$\text{or, } 5 \left[ \frac{2801.7}{10^3} + 2812.5 + 49 \right] + (5 \times 10^3) = 5 \left[ \frac{2089.99}{10^3} + 12800 + 9.8 \right] + \dot{W}_x$$

$$\text{or, } 5 \times 26230.7 + 5000$$

$$= 135139.99 \text{ kW}$$



$$5[280170 + 2812.5 + 49] - 5000 = 5[2089990 + 1280050 + 9.8]$$

$$4022807.5 - 5000 = 10513999 + w_x$$

$$-804.56 \quad 14017807.5 = 10513999 + w_x$$

$$w_x = 3503808.5 \text{ J/s}$$

$$w_x = 3.5 \text{ MW}$$

inlet

$$= 1 \text{ MPa} = 10^6 \text{ Pa} = 10^3 \text{ kPa} = 1000 \text{ kPa}$$

$$= 300^\circ \text{C} + 273 = 573 \text{ K}$$

$$= 30 \text{ m/s}$$

outlet

$$= 0.3 \text{ MPa} = 0.3 \times 10^6 \text{ Pa} = 0.3 \times 10^3 \text{ kPa} = 300 \text{ kPa}$$

Applying SFEE to the nozzle, we have

$$h_1 + g z_1 + \frac{v_1^2}{2} + q = h_2 + g z_2 + \frac{v_2^2}{2} + w_x$$

As the process is adiabatic so,  $q = 0$  and  $w_x = 0$

Neglecting the change in P.E and taking it under mass system, we have,

$$h_1 + \frac{v_1^2}{2} = h_2 + \frac{v_2^2}{2} \quad \text{--- (i)}$$

At inlet : at  $10^3 \text{ kPa} = 1000 \text{ kPa}$

$$t_1 = 179.92^\circ \text{C}$$

Since,  $t_1 > t_{\text{sat}}$  so, the steam is superheated one.

So, from steam table, at  $300^\circ\text{C}$ :-

$$v_g = 0.2579 \text{ m}^3/\text{kg}$$

$$h_g = 3050.6 \text{ kJ/kg} = 3050.6 \times 10^3 \text{ J/kg}$$

$$s_g = 7.1219 \text{ kJ/kgK}$$

At outlet, at  $0.3 \times 10^5 \text{ kPa} = 300 \text{ kPa}$   
 since, the process is adiabatic. So,

$$s_f = s_g = 7.1219 \text{ kJ/kgK}$$

$$\text{Then, } s_f = 1.6721 \text{ kJ/kgK}$$

$$s_{fg} = 5.3200 \text{ kJ/kgK}$$

$$s_g = 6.9921 \text{ kJ/kgK}$$

$$h_f = 561.61 \text{ kJ/kgK}$$

$$h_{fg} = 2163.7 \text{ kJ/kgK}$$

$$h_g = 2725.3 \text{ kJ/kgK}$$

Since,  $s_f < s_g$  so, the steam is superheated one.

$$\therefore h_2 = 2725.3 \text{ kJ/kg}$$

$$\therefore s_2 = s_f + x \cdot s_{fg}$$

$$\text{or, } 7.1219 = 1.6721 + x \cdot 5.3200$$

$$\therefore x = 1.02$$

$$\therefore h_2 = h_f + x \cdot h_{fg} = 561.61 + (1.02 \times 2163.7)$$

$$= 2778.10 \text{ kJ/kgK} = 2778.10 \times 10^3 \text{ J/kgK}$$

$$\text{Then, eqn (i) becomes, } 3050.6 \times 10^3 + \frac{30^2}{2} = \frac{2778.10 \times 10^3}{10^3} + \frac{v_2^2}{2}$$

$$\text{or, } 3050600 + 450 = 2778100 + \frac{v_2^2}{2}$$

$$\text{or, } 3051050 - 2778100 = \frac{v_2^2}{2}$$

$$\text{or, } v_2^2 = 545900$$

$$\therefore v_2 = 738.85 \text{ m/s} //$$

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NAST

BE Computer II  
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